

FACTORING & POLYNOMIALS — Cheatsheet

Algebra 1 · Topics 1–6 · Esme · pre-approved by Mr. Mistry

★ GCF — ALWAYS FIRST STEP

Greatest Common Factor — pull out the biggest shared piece before anything else.

- Find biggest number that divides ALL coefficients.
- Variables: take the SMALLEST power that appears in every term.
- Multiply for the GCF. Pull it out front.
- Divide each term by GCF; remainder goes in parens.
- Check by multiplying back out.

$$6x^3 + 9x^2 = 3x^2(2x + 3)$$

Trap: when a term EQUALS the GCF, it leaves 1 (not 0).

Factor $x^2 + bx + c$ ($a = 1$)

Find two numbers that multiply to c AND add to b.

- Check GCF first.
- List factor pairs of c.
- Find the pair that adds to b.
- Write $(x + p)(x + q)$.
- Check by FOIL.

$$x^2 + 7x + 12 \rightarrow 3 \cdot 4 = 12, \quad 3 + 4 = 7 \rightarrow (x+3)(x+4)$$

c sign	b sign	p, q signs
+	+	both +
+	-	both -
-	any	opposite signs; bigger one matches b's sign

Factor $ax^2 + bx + c$ — AC Method

When the leading coefficient isn't 1. A-C, F-S-G-P.

- Check GCF first.
- Multiply $a \cdot c$.
- Find two #s that $x=ac$ AND $=b$.
- Split bx into those two terms.
- Factor by grouping (pair up, pull out GCF of each pair).
- Pull out the shared parens.

$$\begin{aligned} 2x^2 + 7x + 3 \\ ac=6, \text{ find 6\&1 } (x6, +7) \\ 2x^2 + 6x + x + 3 \\ 2x(x+3) + 1(x+3) \\ (2x+1)(x+3) \end{aligned}$$

If groups don't share parens, try other sign combo.

Difference of Squares

Two perfect squares separated by a MINUS.

$$a^2 - b^2 = (a + b)(a - b)$$

- Check GCF first.
- Confirm: 2 terms, both perfect squares, minus between.
- Identify a and b (square roots).
- Write $(a + b)(a - b)$.

$$\begin{aligned} x^2 - 25 &= (x+5)(x-5) \\ 4x^2 - 9 &= (2x+3)(2x-3) \\ x^4 - 16 &= (x^2+4)(x^2-4) = (x^2+4)(x+2)(x-2) \end{aligned}$$

Sum of squares (a^2+b^2) does NOT factor.

FOIL — Multiply Two Binomials

First · Outer · Inner · Last. Use to check factoring.

$$(a+b)(c+d) = ac + ad + bc + bd$$

$$\begin{aligned} (x+4)(x-3) &= x^2 - 3x + 4x - 12 = \\ &= x^2 + x - 12 \end{aligned}$$

Special Products

Pattern	Expansion
$(a+b)^2$	$a^2 + 2ab + b^2$
$(a-b)^2$	$a^2 - 2ab + b^2$
$(a+b)(a-b)$	$a^2 - b^2$

Trap: $(x+3)^2 \neq x^2 + 9$. Middle 2ab is REQUIRED!

Add / Subtract Polynomials

- For subtraction: **FLIP every sign** in the 2nd polynomial.
- Drop parentheses.
- Combine LIKE TERMS (same var + same exponent).
- Write in standard form (biggest exponent first).

$$\begin{aligned} (3x^2+5x-2) - (x^2-4x+7) \\ = (3x^2+5x-2) + (-x^2+4x-7) \\ = 2x^2 + 9x - 9 \end{aligned}$$

Every sign flips — including the constant!

Perfect Squares Lookup

n	1	2	3	4	5	6	7	8	9
n ²	1	4	9	16	25	36	49	64	81
n	11	12	13	14	15	16	20	25	30
n ²	121	144	169	196	225	256	400	625	900
								100	10000

$x^2, 4x^2, 9x^2, 16x^2, 25x^2 \dots$ are all perfect squares.
(Coefficient must be a perfect square AND the variable's exponent must be even.)

★ FACTORING DECISION TREE

- GCF?** Always pull it out first.
- 2 terms?** Check Difference of Squares (a^2-b^2). Sum of squares = stuck.
- 3 terms, $a=1$?** Find p,q where $pq=c$, $p+q=b$.
- 3 terms, $a \neq 1$?** AC method (multiply a-c, find #s, split, group, pull).
- 4 terms?** Factor by grouping (pair up).
- Stuck?** Try: rearrange, look for hidden GCF, or it's prime.

Sign Rules (multiplication)

a	b	a-b
+	+	+
-	-	+
+	-	-
-	+	-

Same signs $\rightarrow +$. Different $\rightarrow -$.

Vocab Quick-Ref

Polynomial	Sum of terms with variables
Monomial	1 term ($3x^2$)
Binomial	2 terms ($x+3$)
Trinomial	3 terms (x^2+5x+6)
Coefficient	Number in front (5 in $5x$)
Constant	Plain number, no variable
Like terms	Same var + same exp
Degree	Biggest exponent
Standard form (poly)	Biggest exp first
Factor (verb)	Un-do FOIL
Prime polynomial	Won't factor (over integers)

⚠ TOP 3 FACTORING TRAPS

- Forgetting GCF.** Always step 1. Free easier numbers.
- $(x+3)^2 \neq x^2+9$.** Middle 2ab term required. Write it twice: $(x+3)(x+3)$.
- Sum of squares.** a^2+b^2 does NOT factor in Algebra 1. Walk away.